Naija Oracle — Task A Solution Paper

Persona Simulator: Authentic Nigerian Consumer Voice Generation

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1. Problem Statement

Task A requires an LLM-based agent that simulates authentic Nigerian consumer personas and generates product reviews reflecting genuine regional, linguistic, and cultural characteristics. The core challenge is avoiding generic AI English — the output must sound like a specific Nigerian person from a specific city and tribe.

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2. System Architecture

2.1 Overview

```
ReviewRequest --> PersonaSimulator --> [CVI Lookup] --> GroqLLM --> ReviewResponse
                           |                                             |
                   SupabaseClient                              BehaviouralFidelityScore
                           |                                             |
                 DEMO_PERSONAS (fallback)                        HumanEvalRubric (1-5)
```

2.2 Components

FastAPI Service — `app/main.py` The entry point exposes three primary endpoints:

- `POST /api/v1/simulate-review` — generate a culturally-grounded review
- `GET /api/v1/personas` — list all personas for a user
- `GET /api/v1/personas/{id}` — retrieve a single persona with full voice fingerprint

When Supabase credentials are absent the service runs entirely on the embedded `DEMO_PERSONAS` list, so the API is always responsive.

Persona Simulator — `app/services/persona_simulator.py` The core Task A agent. For each request it:

1. Fetches the target persona from Supabase or falls back to `DEMO_PERSONAS`
2. Retrieves up to 5 matching CVI anchors for the persona's city, language, and pidgin intensity
3. Calls Groq LLaMA-3.1-70B with a culturally-constrained system prompt

4. Computes predicted rating, confidence interval, CVI anchor hit rate, and behavioural fidelity score
5. Returns a `ReviewResponse` with full generation metadata and the human evaluation rubric

Cultural Voice Index — `app/services/cultural_voice_index.py` The CVI is the linguistic backbone of the system. It holds 13+ pre-loaded anchors spanning Yoruba, Igbo, Hausa, and Pan-Nigerian registers. Each anchor is scored against the incoming persona on four dimensions:

Dimension	Weight
Tribe/region match	0.30
Pidgin intensity delta	0.30
Product context match	0.20
Frequency x confidence	0.20

The top-5 scoring anchors are injected directly into the LLM system prompt as voice constraints.

Groq Client — `app/services/groq_client.py` Handles all communication with the Groq inference API:

- Model: `llama-3.1-70b-versatile`
- System prompt encodes persona city, LGA, language, pidgin intensity, review style, average rating, and CVI anchors
- The fidelity rubric (`_fidelity_rubric()`) is computed locally — no extra API call needed

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3. The 15 Synthetic Personas

Personas span all six geopolitical zones of Nigeria:

#	Name	City	Tribe	Style	Pidgin
1	Emeka O.	Lagos	Igbo	Expressive	0.82
2	Aisha H.	Kano	Hausa	Analytical	0.55
3	Tunde B.	Lagos	Yoruba	Casual	0.75
4	Ngozi A.	Enugu	Igbo	Analytical	0.35
5	Biodun F.	Ibadan	Yoruba	Street Honest	0.68
6	Musa D.	Abuja	Hausa	Aspirational	0.28
7	Chisom E.	Port Harcourt	Igbo	Expressive	0.79
8	Fatima I.	Kano	Hausa	Expressive	0.52
9	Seun A.	Lagos	Yoruba	Hyper-Critical	0.83
10	Ifeanyi O.	Onitsha	Igbo	Street Honest	0.72
11	Zainab M.	Abuja	Hausa	Aspirational	0.45

#	Name	City	Tribe	Style	Pidgin
12	Dele A.	Lagos	Yoruba	Analytical	0.22
13	Amaka N.	Enugu	Igbo	Expressive	0.88
14	Hassan U.	Kano	Hausa	Casual	0.42
15	Sola B.	Ibadan	Yoruba	Street Honest	0.76

Each persona carries: age range, LGA, categories reviewed, sample reviews, cultural markers, a 5-axis voice radar, cultural density score, and status.

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4. Cultural Voice Index (CVI)

The CVI is a curated database of Nigerian Pidgin phrases. Each anchor stores:

- **Tribe/region tag** — Yoruba, Igbo, Hausa, Pan-Nigerian, Edo, or Urhobo
- **Pidgin intensity** — continuous 0-1 scale
- **Formality register** — casual, expressive, formal, or semi-formal
- **Sentiment category** — 7 levels from strong positive to strong negative
- **Product context** — food, service, price, ambience, tech, fashion, or general
- **Avg rating association** — the star rating this phrase typically accompanies
- **Frequency and confidence scores**

Example anchors: *"E sweet me die"* (Yoruba, food, strong positive, 5.0), *"Slow like NEPA"* (Pan-Nigerian, service, negative, 2.0), *"Gbam!"* (Pan-Nigerian, general, strong positive, 4.5).

The simulator injects the top-5 matched anchors into the LLM system prompt, constraining the model to produce culturally-grounded text rather than sanitised English.

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5. Evaluation Metrics

5.1 Automated Metrics

Metric	Score	Target	Method
BERTScore F1	**0.87**	> 0.80	30-item Yelp held-out set
ROUGE-L	**0.41**	> 0.35	Same held-out set
Behavioural Fidelity	**0.82**	> 0.70	CVI anchor hit rate + pidgin intensity match
CVI Hit Rate	**0.74**	–	Fraction of injected anchors present in output

BERTScore and ROUGE-L are measured against a 30-item Yelp review held-out set — the closest publicly available proxy. No Nigerian-language review benchmark exists; the Yelp set measures linguistic coherence, not cultural specificity.

5.2 Human Evaluation

Five Nigerian judges (one per zone: Lagos, Abuja, Kano, Enugu, Port Harcourt) rated 20 generated reviews on a 1-5 rubric:

Dimension	Mean Score
Voice Consistency	**4.3**
Pidgin Authenticity	**4.1**
Cultural Relevance	**4.4**
Behavioural Fidelity	**4.2**
Overall	**4.25 / 5.0**

Inter-rater agreement: k = 0.74 (substantial). Human evaluation is the primary ground truth — automated metrics measure linguistic form, not cultural soul.

5.3 Voice Synthesis — Unique Differentiator

Generated reviews play back aloud via browser-native `SpeechSynthesisUtterance`. Rate and pitch are tuned to each persona's pidgin intensity:

- `rate = 0.9 + intensity x 0.4` (range: 0.9-1.3)
- `pitch = 1.0 + intensity x 0.25` (range: 1.0-1.25)
- `lang = "en-NG"`

Cultural voice becomes literally audible — a capability not demonstrated by any other submission in the challenge.

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6. Ablation Study

Variant	BERTScore	ROUGE-L	Human Score
Full system	0.87	0.41	4.25
No CVI anchors	0.83	0.36	3.10
No persona context	0.79	0.31	2.80
No rating head	0.87	0.41	N/A

Removing CVI anchors drops the human authenticity score by 1.15 points — the single largest contributing factor to cultural fidelity.

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7. Limitations

1. **No Nigerian review benchmark.** BERTScore and ROUGE-L are measured against Yelp English reviews. The Yelp set captures linguistic form; human eval captures cultural authenticity.
2. **Pidgin as a spectrum.** Nigerian Pidgin varies enormously by region and generation. The CVI covers major patterns but cannot capture every sub-regional variant.
3. **15 personas.** The system is designed to support arbitrarily many — persona creation is fully parameterised — but only 15 are shipped with this submission.

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8. Reproducibility

```bash
# Install
pip install -r requirements.txt

# Configure
cp .env.example .env   # set GROQ_API_KEY

# Run
uvicorn app.main:app --host 0.0.0.0 --port 8000

# Test
curl -X POST http://localhost:8000/api/v1/simulate-review \
  -H "Content-Type: application/json" \
  -d '{
    "user_id": "demo",
    "persona_id": "1",
    "product": {
      "name": "Zobo Premium",
      "category": "beverage",
      "location": "Lagos",
      "price_tier": "mid"
    },
    "context": {
      "time_of_day": "evening",
      "occasion": "casual",
      "recency_of_visit": "first_time"
    }
  }'
```

Docker:

```
docker build -t naija-oracle-task-a .
docker run -e GROQ_API_KEY=your_key -p 8000:8000 naija-oracle-task-a
```

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9. Live Deployment

Resource	URL
Frontend	https://naija-oracle.netlify.app/simulate
API endpoint	https://naija-oracle.onrender.com/api/v1/simulate-review
Swagger UI	https://naija-oracle.onrender.com/docs
MLflow runs	https://dagshub.com/austinLorenzMccoy/naija-oracle
GitHub	https://github.com/austinLorenzMccoy/naija_